

Live Sport Streaming Analysis

CS342: Computer Networks Lab

Group No.: 47

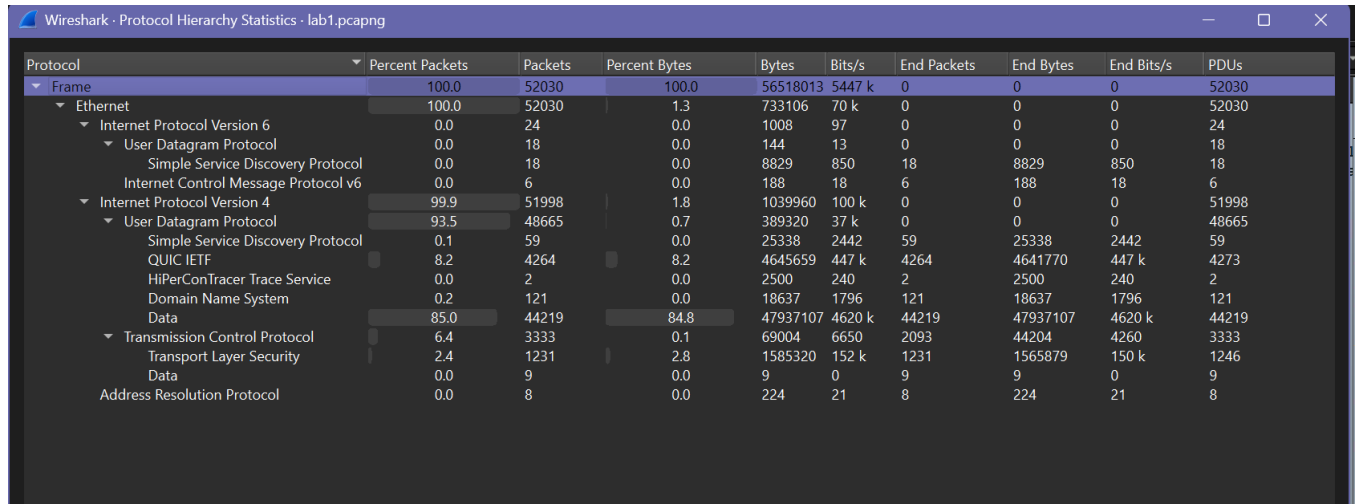
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1 Protocols Observed & Packet Formats (Q a)

From traces of live sport streaming using the hotstar application, the following protocols were seen:



The image shows a screenshot of the Wireshark Protocol Hierarchy Statistics window for a file named lab1.pcapng. The window displays a tree view of protocols and their corresponding statistics. The 'Frame' protocol is at the top, followed by 'Ethernet', 'Internet Protocol Version 6', 'User Datagram Protocol', 'Simple Service Discovery Protocol', 'Internet Control Message Protocol v6', 'Internet Protocol Version 4', 'User Datagram Protocol', 'Simple Service Discovery Protocol', 'QUIC IETF', 'HiPerConTracer Trace Service', 'Domain Name System', 'Data', 'Transmission Control Protocol', 'Transport Layer Security', 'Data', and 'Address Resolution Protocol'. The statistics for each protocol are shown in a table with columns: Protocol, Percent Packets, Packets, Percent Bytes, Bytes, Bits/s, End Packets, End Bytes, End Bits/s, and PDUs.

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	52030	100.0	56518013	5447 k	0	0	0	52030
Ethernet	100.0	52030	1.3	733106	70 k	0	0	0	52030
Internet Protocol Version 6	0.0	24	0.0	1008	97	0	0	0	24
User Datagram Protocol	0.0	18	0.0	144	13	0	0	0	18
Simple Service Discovery Protocol	0.0	18	0.0	8829	850	18	8829	850	18
Internet Control Message Protocol v6	0.0	6	0.0	188	18	6	188	18	6
Internet Protocol Version 4	99.9	51998	1.8	1039960	100 k	0	0	0	51998
User Datagram Protocol	93.5	48665	0.7	389320	37 k	0	0	0	48665
Simple Service Discovery Protocol	0.1	59	0.0	25338	2442	59	25338	2442	59
QUIC IETF	8.2	4264	8.2	4645659	447 k	4264	4641770	447 k	4273
HiPerConTracer Trace Service	0.0	2	0.0	2500	240	2	2500	240	2
Domain Name System	0.2	121	0.0	18637	1796	121	18637	1796	121
Data	85.0	44219	84.8	47937107	4620 k	44219	47937107	4620 k	44219
Transmission Control Protocol	6.4	3333	0.1	69004	6650	2093	44204	4260	3333
Transport Layer Security	2.4	1231	2.8	1585320	152 k	1231	1565879	150 k	1246
Data	0.0	9	0.0	9	0	9	9	0	9
Address Resolution Protocol	0.0	8	0.0	224	21	8	224	21	8

- **Ethernet (L2)**
- **IP (L3)**
- **TCP (L4)** – Src/Dst port (443 for HTTPS), SEQ/ACK, flags.
- **TLS (Security)**
- **DNS** – Resolutions of Hotstar CDN domains.

Each protocol encapsulates the upper layer and enables reliable, secure streaming.

Ethernet (Layer 2)

Dest MAC	Src MAC	EtherType
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IPv4 (Layer 3)

Ver	IHL	Total Len	TTL
Protocol	Src IP Addr	Dst IP	

TCP (Layer 4)

+-----+-----+	
Src Port	Dst Port
+-----+-----+	
Sequence Number	
+-----+-----+	
Acknowledgment Number	
+-----+-----+	
Flags	Window, Checksum ...
+-----+-----+	

TLS Handshake

Content Type	Version	Length
Handshake Type	Cipher Suites	SNI...

2 Observed Field Values (Q b)

Field values from the traces are shown below:

```
▼ Ethernet II, Src: Client (28:cd:c4:1d:c4:0f), Dst: Gateway (30:de:4b:c7:2d:02)
  ► Destination: Gateway (30:de:4b:c7:2d:02)
  ► Source: Client (28:cd:c4:1d:c4:0f)
    Type: IPv4 (0x0800)
    [Stream index: 0]

▼ Domain Name System (response)
  Transaction ID: 0x51cf
  ► Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 1
  Authority RRs: 4
  Additional RRs: 2
  ▼ Queries
    ► aba5c6eaaaaaaaamon3ym4l56kplk.live-cf.cdn.hotstar.com: type A, class IN
  ▼ Answers
    ► aba5c6eaaaaaaaamon3ym4l56kplk.live-cf.cdn.hotstar.com: type A, class IN, addr 70.130.228.70
  ▼ Authoritative nameservers
    ► hotstar.com: type NS, class IN, ns ns-484.awsdns-60.com
    ► hotstar.com: type NS, class IN, ns ns-1645.awsdns-13.co.uk
    ► hotstar.com: type NS, class IN, ns ns-753.awsdns-30.net
    ► hotstar.com: type NS, class IN, ns ns-1296.awsdns-34.org
  ▼ Additional records
    ► ns-753.awsdns-30.net: type A, class IN, addr 205.251.194.241
    ► ns-484.awsdns-60.com: type A, class IN, addr 205.251.193.228
    [Request In: 129]
    [Time: 0.057371000 seconds]
```

dns.qry.name contains "hotstar"							
No.	Time	Source	Destination	Protocol	Length	Info	
116	20:23:43.097359	Client	Gateway	DNS	83	Standard query	0x81df A bifrost-api.hotstar.com
117	20:23:43.097585	Client	Gateway	DNS	83	Standard query	0x18bb HTTPS bifrost-api.hotstar.com
118	20:23:43.098789	Gateway	Client	DNS	529	Standard query response	0x81df A bifrost-api.hotstar.com CNAME bifrost-api.h
119	20:23:43.099885	Gateway	Client	DNS	241	Standard query response	0x18bb HTTPS bifrost-api.hotstar.com CNAME bifrost-a
129	20:23:43.134484	Client	Gateway	DNS	113	Standard query	0x51cf A aba5c6eaaaaaaaamon3ym4l56kplk.live-cf.cdn.hotstar.c
130	20:23:43.134785	Client	Gateway	DNS	113	Standard query	0x08af HTTPS aba5c6eaaaaaaaamon3ym4l56kplk.live-cf.cdn.hotsta
136	20:23:43.166082	Gateway	Client	DNS	194	Standard query response	0x08af HTTPS aba5c6eaaaaaaaamon3ym4l56kplk.live-cf.c
138	20:23:43.191855	Gateway	Client	DNS	298	Standard query response	0x51cf A aba5c6eaaaaaaaamon3ym4l56kplk.live-cf.cdn.f

```
▼ Internet Protocol Version 4, Src: Server_hotstar (70.130.228.70), Dst: Client (192.168.1.102)
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ► Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 1480
    Identification: 0x4a7b (19067)
  ► 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 241
    Protocol: TCP (6)
    Header Checksum: 0x8cdd [validation disabled]
    [Header checksum status: Unverified]
    Source Address: Server_hotstar (70.130.228.70)
    Destination Address: Client (192.168.1.102)
    [Stream index: 21]
```

```
▼ Transmission Control Protocol, Src Port: 55472 (55472), Dst Port: https (443), Seq: 4811, Ack: 159693, Len: 0
  Source Port: 55472 (55472)
  Destination Port: https (443)
  [Stream index: 16]
  [Stream Packet Number: 160]
  ► [Conversation completeness: Incomplete, DATA (15)]
  [TCP Segment Len: 0]
  Sequence Number: 4811 (relative sequence number)
  Sequence Number (raw): 134479176
```

3 Message Sequences & Handshakes (Q c)

The streaming session follows this sequence:

1. **DNS lookup** for Hotstar domains.
2. **TCP 3-way handshake** (SYN, SYN-ACK, ACK).
3. **TLS handshake** (ClientHello with ServerHello, certificate, key exchange).
4. **Segment downloads** – Repeated requests for small chunks.

No.	Time	Source	Destination	Protocol	Length	Info
125	20:23:43.105610	Client	bifrost-api.hotstar.com	TLSv1.3	662	Client Hello (SNI=bifrost-api.hotstar.com)
144	20:23:43.259980	Client	Server_hotstar	TLSv1.3	377	Client Hello (SNI=abasc5eaaaaaamon3ym4id56kpl.live-cf.cdn.hotstar.com)
147	20:23:43.324146	Server_hotstar	Client	TLSv1.3	1494	Server Hello, Change Cipher Spec, Application Data
175	20:23:43.434634	bifrost-api.hotstar.com	Client	TLSv1.3	308	Server Hello, Change Cipher Spec, Application Data, Application Data
229	20:23:43.860566	Client	www.google.co.in	QUIC	1292	Initial, DCID=7518363e6e08ceb5, PKN: 2, PING, CRYPTO, CRYPTO, CRYPTO, CRYPTO, CRYPTO
267	20:23:43.958581	www.google.co.in	Client	QUIC	1292	Initial, SCID=f518363e6e08ceb5, PKN: 3, CRYPTO, PADDING
6128	20:23:51.382675	Client	ma.hsr.im	TLSv1.3	409	Client Hello (SNI=ma.hsr.im)
6132	20:23:51.845372	ma.hsr.im	Client	TLSv1.3	2974	Server Hello, Change Cipher Spec, Application Data
7172	20:23:54.357552	Client	www.bing.com	QUIC	1292	Initial, DCID=7aa8f93af30d51b7, PKN: 2, CRYPTO, PING, CRYPTO, PADDING, CRYPTO
7175	20:23:54.644251	www.bing.com	Client	QUIC	1292	Initial, SCID=008edea1064d8624, PKN: 3, CRYPTO, PADDING
7212	20:23:59.744713	Client	www.google.com	QUIC	1292	Initial, DCID=a6c3837f473520f6, PKN: 2, PING, CRYPTO, PING, PING, PING, CRYPTO
7217	20:24:00.076353	www.google.com	Client	QUIC	1292	Initial, SCID=c6c3837f473520f6, PKN: 3, CRYPTO, PADDING
11473	20:24:15.110806	Client	remotedesktop-pa.googleleapi...	QUIC	1292	Initial, DCID=57fb94461e325fb2, PKN: 2, PING, PADDING, PING, CRYPTO, PADDING
11477	20:24:15.208723	remotedesktop-pa.googleleapi...	Client	QUIC	1292	Initial, SCID=f7fb94461e325fb2, PKN: 3, CRYPTO, PADDING
12321	20:24:24.956267	Client	beacons.gcp.gvt2.com	QUIC	1292	Initial, DCID=c865f87ff9dc26c4, PKN: 2, CRYPTO, PING, CRYPTO, CRYPTO, CRYPTO, CRYPTO
12326	20:24:25.062676	beacons.gcp.gvt2.com	Client	QUIC	1292	Initial, SCID=c865f87ff9dc26c4, PKN: 3, CRYPTO, PADDING
14359	20:24:33.383263	Client	q.quora.com	QUIC	1292	Initial, DCID=554404eb360dce71, PKN: 2, PADDING, CRYPTO, PING, PADDING, CRYPTO
14367	20:24:33.429101	q.quora.com	Client	QUIC	1242	Initial, SCID=01db5b330dd9a463e6d9fd3370d9a928d5da4d1e, PKN: 2, CRYPTO
14368	20:24:33.429101	q.quora.com	Client	QUIC	1242	Handshake, SCID=01db5b330dd9a463e6d9fd3370d9a928d5da4d1e
15475	20:24:34.995247	Client	google.com	QUIC	1292	Initial, DCID=ee845f38a75b41a5, PKN: 2, PADDING, CRYPTO, PADDING, PING, PING
15479	20:24:35.195874	google.com	Client	QUIC	1292	Initial, SCID=ee845f38a75b41a5, PKN: 3, CRYPTO, PADDING

- Transport Layer Security
 - TLSv1.3 Record Layer: Handshake Protocol: Client Hello

120	20:23:43.100338	Client	bifrost-api.hotstar.com	TCP	66	6278 → https(443)	[SYN]	Seq=0	Win=65535	Len=0	MSS=1460	WS=256	SACK_PERM
122	20:23:43.104898	bifrost-api.hotstar.com	Client	TCP	66	https(443) → 6278	[SYN, ACK]	Seq=0	Ack=1	Win=18352	Len=0	MSS=1460	SACK_PERM
139	20:23:43.192825	Server_hotstar	Server_hotstar	TCP	66	55472 → https(443)	[SYN]	Seq=0	Win=65535	Len=0	MSS=1460	WS=256	SACK_PERM
141	20:23:43.259196	Client	Client	TCP	66	https(443) → 55472	[SYN, ACK]	Seq=0	Ack=1	Win=65535	Len=0	MSS=1440	SACK_PERM

No.	Time	Source	Destination	Protocol	Length	Info
125	20:23:43.105610	Client	bifrost-api.hotstar.com	TLSv1.3	662	Client Hello (SNI=bifrost-api.hotstar.com)
144	20:23:43.259980	Client	Server.hotstar	TLSv1.3	377	Client Hello (SNI=aba5c6eaaaaaaaam0n3ym4l56kplk.live-cf.cdn.hotstar.com)
147	20:23:43.324146	Server.hotstar	Client	TLSv1.3	1494	Server Hello, Change Cipher Spec, Application Data
175	20:23:43.434634	bifrost-api.hotstar.com	Client	TLSv1.3	308	Server Hello, Change Cipher Spec, Application Data, Application Data

4 Protocol Relevance to Application (Q d)

- **Ethernet/IP** – Basic delivery over LAN/Internet.
- **TCP** – Reliable transport for video data.
- **TLS** – End-to-end encryption for security.
- **DNS** – Maps Hotstar domains to nearest CDN servers.

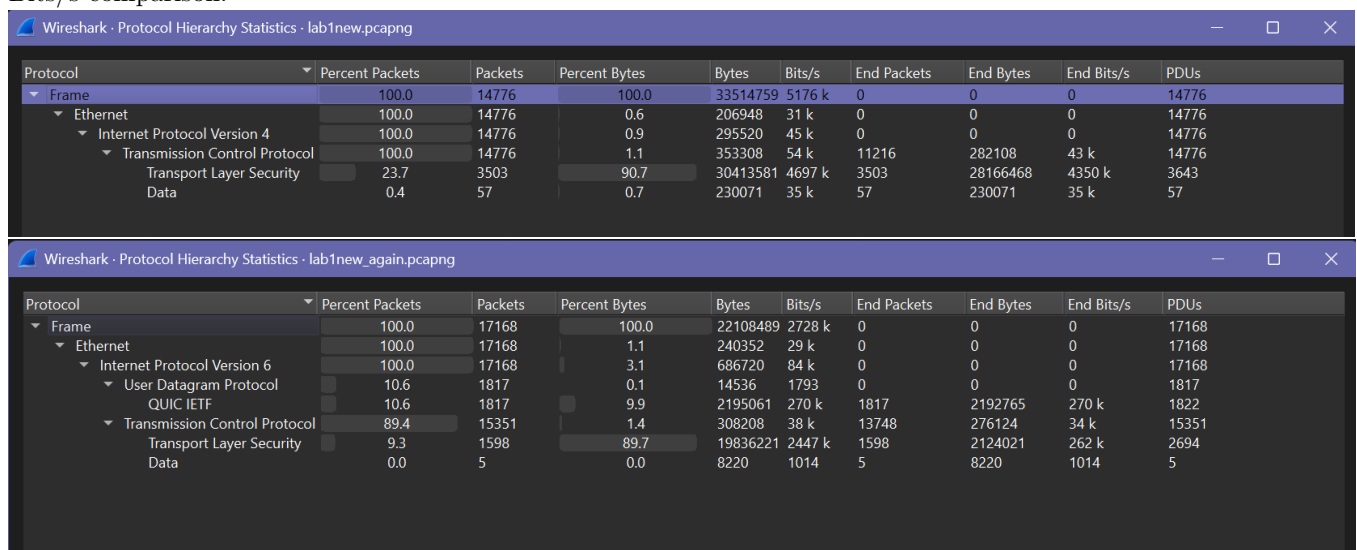
These protocols together ensure secure, adaptive, and low-latency live sports streaming.

5 Time-of-Day Statistics (Q e)

To understand variation in network performance, traces were captured at different times of the day. Metrics such as throughput, RTT, packet size, loss percentage were measured.

Results

Bits/s comparison:

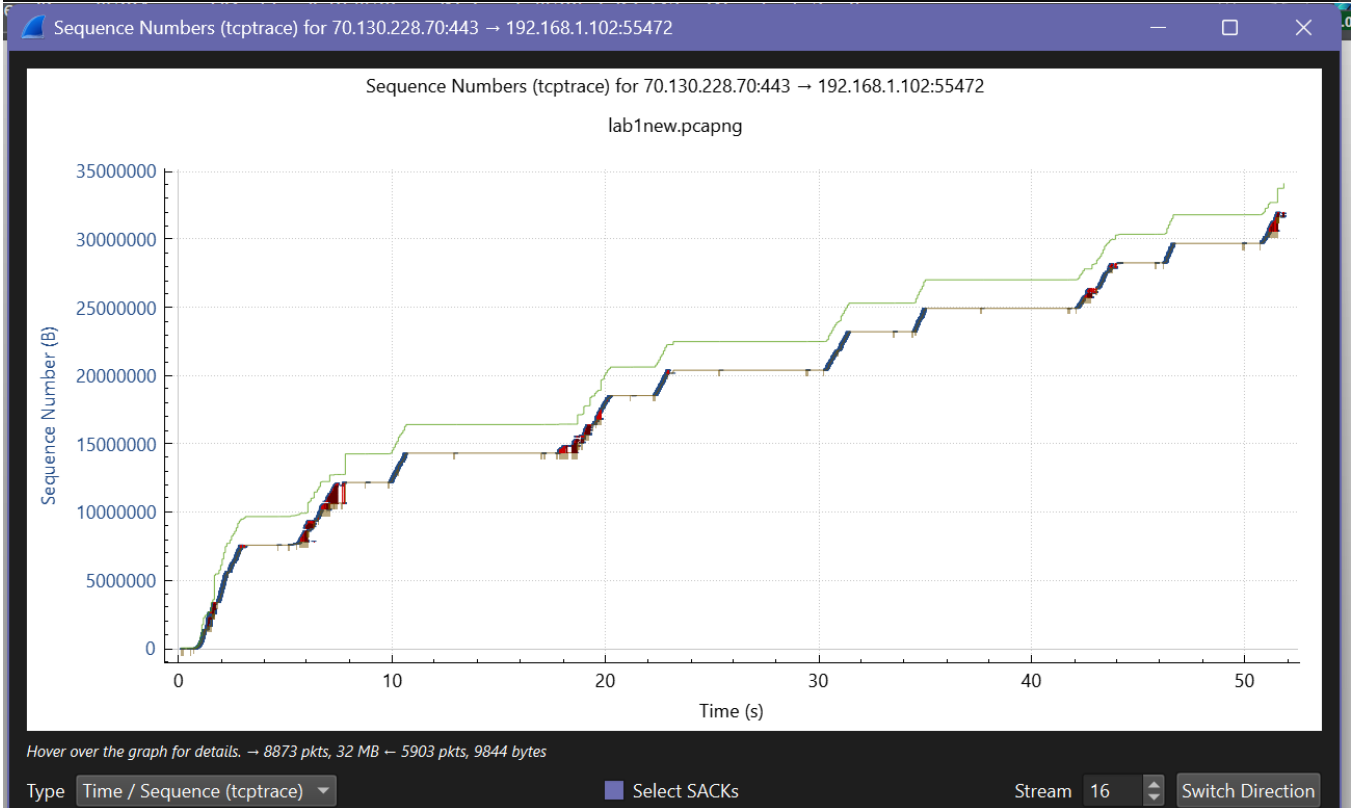
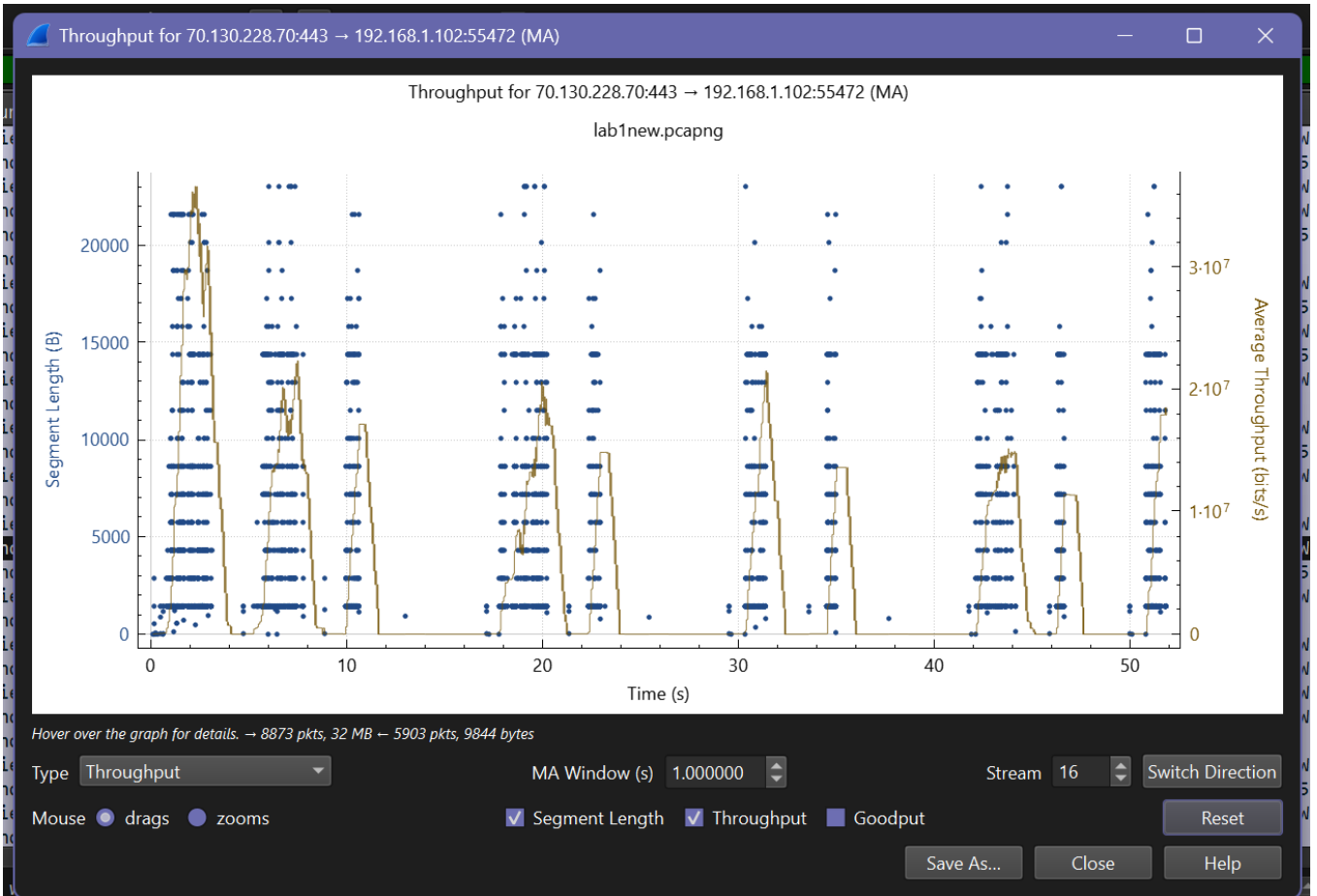


The image displays two screenshots of the Wireshark Protocol Hierarchy Statistics window. The top screenshot is for 'lab1new.pcapng' and the bottom is for 'lab1new_again.pcapng'. Both show a tree view of protocols with corresponding statistics for packets, bytes, and bits per second.

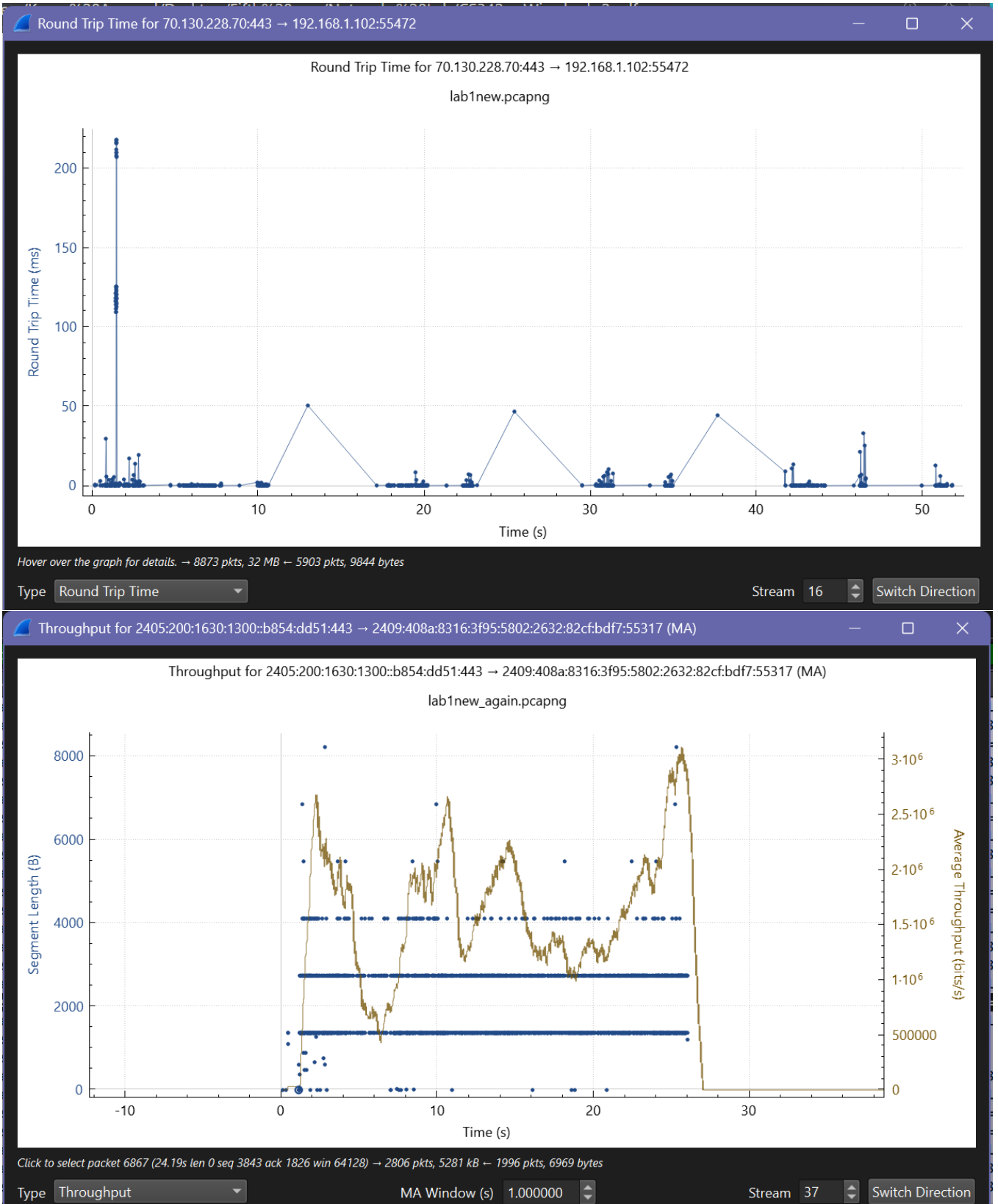
Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	14776	100.0	33514759	5176 k	0	0	0	14776
Ethernet	100.0	14776	0.6	206948	31 k	0	0	0	14776
Internet Protocol Version 4	100.0	14776	0.9	295520	45 k	0	0	0	14776
Transmission Control Protocol	100.0	14776	1.1	353308	54 k	11216	282108	43 k	14776
Transport Layer Security	23.7	3503	90.7	30413581	4697 k	3503	28166468	4350 k	3643
Data	0.4	57	0.7	230071	35 k	57	230071	35 k	57

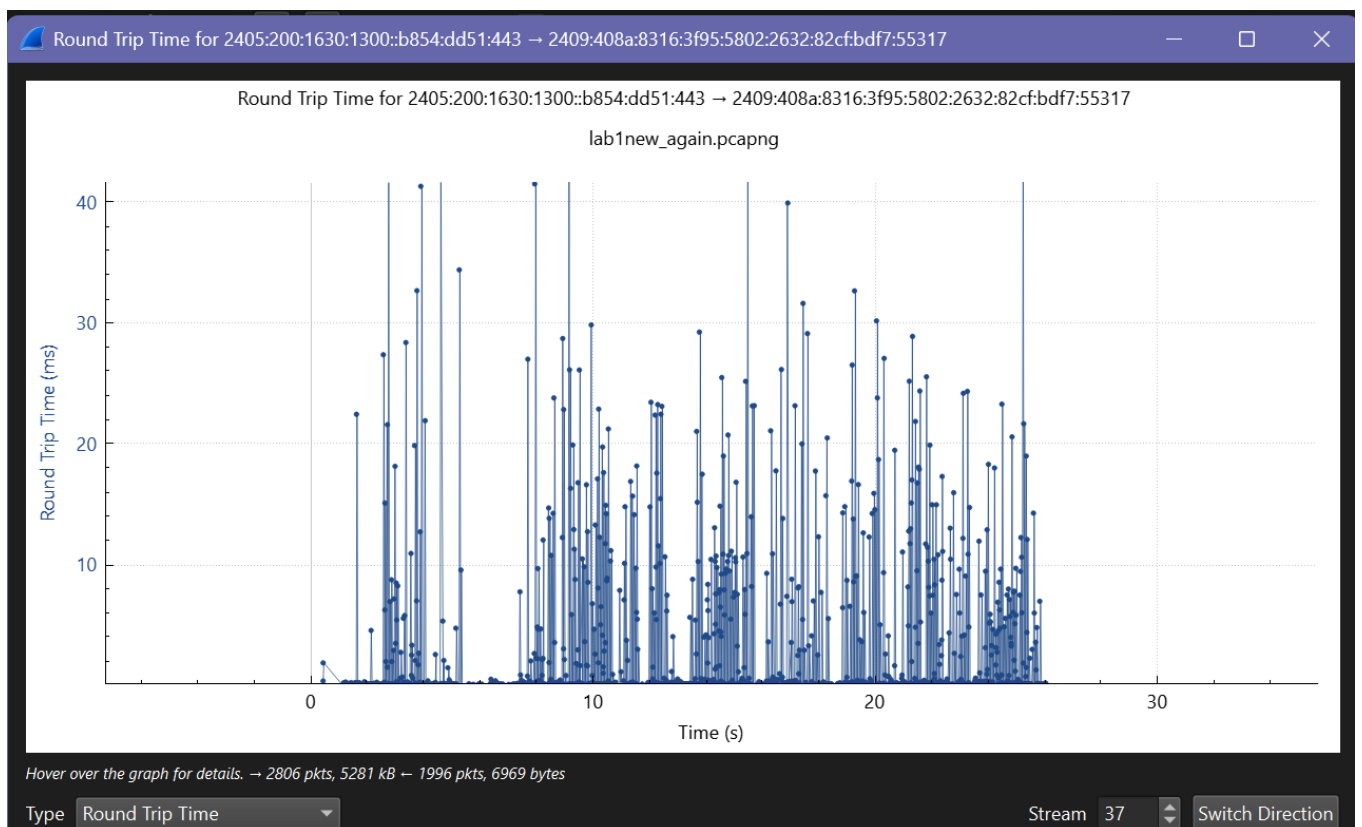
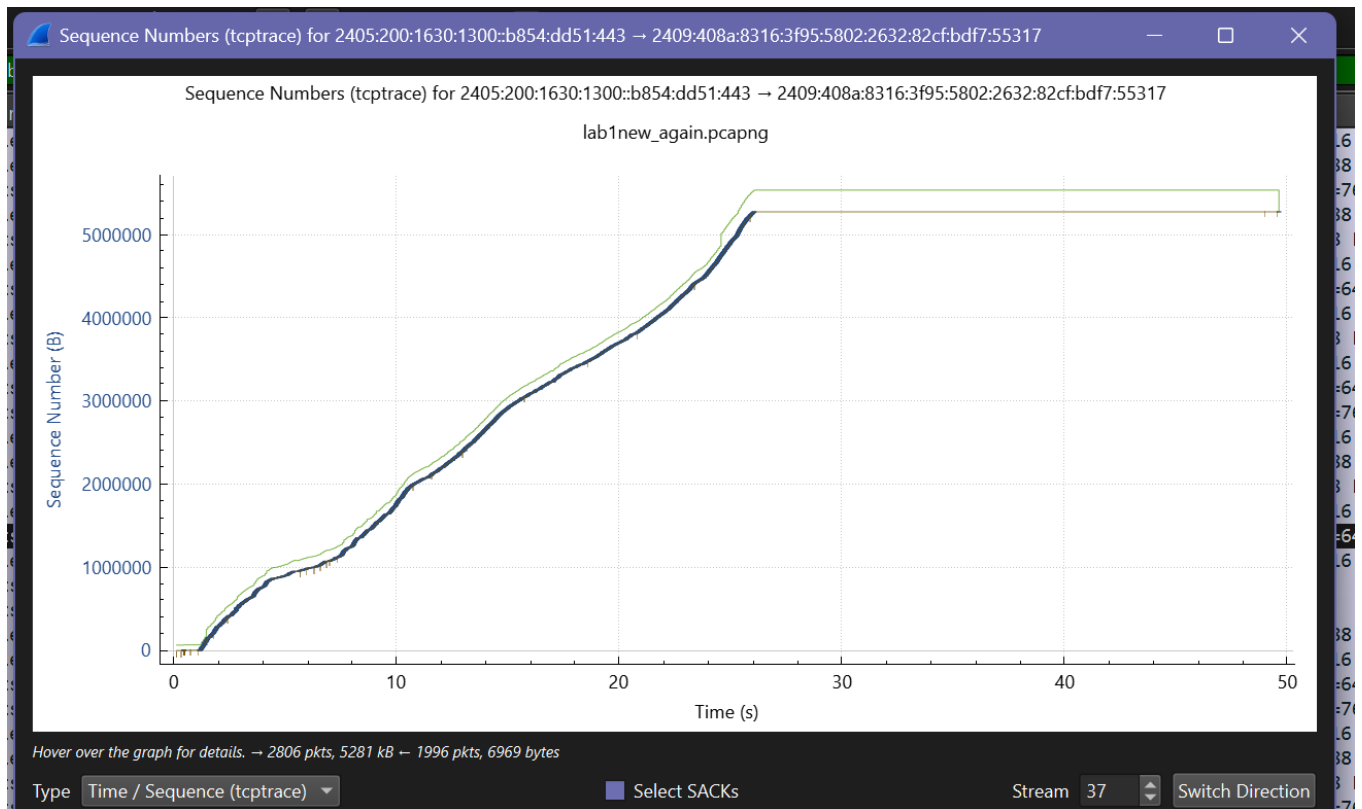
Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	17168	100.0	22108489	2728 k	0	0	0	17168
Ethernet	100.0	17168	1.1	240352	29 k	0	0	0	17168
Internet Protocol Version 6	100.0	17168	3.1	686720	84 k	0	0	0	17168
User Datagram Protocol	10.6	1817	0.1	14536	1793	0	0	0	1817
QUIC IETF	10.6	1817	9.9	2195061	270 k	1817	2192765	270 k	1822
Transmission Control Protocol	89.4	15351	1.4	308208	38 k	13748	276124	34 k	15351
Transport Layer Security	9.3	1598	89.7	19836221	2447 k	1598	2124021	262 k	2694
Data	0.0	5	0.0	8220	1014	5	8220	1014	5

Throughput, tcptrace, RTT comparison:



Bits/s comparison:





Packet Lengths:

Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
Packet Lengths	15500	2178.46	42	23094	0.2572	100%	3.5600	8.649
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	5305	59.59	42	79	0.0880	34.23%	1.1700	58.440
80-159	1074	89.81	80	158	0.0178	6.93%	1.7400	8.690
160-319	76	224.11	160	312	0.0013	0.49%	0.0400	7.812
320-639	64	426.12	321	632	0.0011	0.41%	0.0500	8.003
640-1279	65	1017.37	659	1278	0.0011	0.42%	0.0500	6.952
1280-2559	6108	1492.85	1288	1514	0.1014	39.41%	1.5400	8.397
2560-5119	788	3484.10	2934	4374	0.0131	5.08%	0.4300	9.692
5120 and greater	2020	10583.82	5814	23094	0.0335	13.03%	0.5500	8.876

Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
Packet Lengths	24009	1151.35	42	13774	0.3103	100%	1.7000	10.784
0-19	0	-	-	-	0.0000	0.00%	-	-
20-39	0	-	-	-	0.0000	0.00%	-	-
40-79	8127	73.96	42	79	0.1050	33.85%	0.5100	24.225
80-159	1977	99.60	80	158	0.0256	8.23%	0.6400	10.690
160-319	401	219.35	161	319	0.0052	1.67%	0.1200	9.759
320-639	231	454.51	321	639	0.0030	0.96%	0.0700	10.177
640-1279	249	943.40	640	1262	0.0032	1.04%	0.1100	7.540
1280-2559	8870	1442.13	1281	1514	0.1146	36.94%	1.0600	10.784
2560-5119	4096	3218.04	2814	4184	0.0529	17.06%	0.3800	63.490
5120 and greater	58	7656.24	5554	13774	0.0007	0.24%	0.0700	17.594

netstat -s:

No.	Time	Source	Destination	Protocol	Length	Info
61	20:23:42.575374	Client	123.35.104.34.googleuse...	TCP	66	[TCP Retransmission] 55873 → https(443) [SYN] Seq=0 Win=65535 Len=0 MS
180	20:23:43.445330	Client	bifrost-api.hotstar.com	TCP	1514	[TCP Retransmission] 62787 → https(443) [PSH, ACK] Seq=30888 Ack=255 Wi
1057	20:23:44.515368	Sender_hotstar	Client	TCP	1494	[TCP Previous segment not captured] https(443) → 55472 [ACK] Seq=16596
1062	20:23:44.517344	Sender_hotstar	Client	TCP	1494	[TCP Previous segment not captured] https(443) → 55472 [ACK] Seq=16746
1063	20:23:44.517344	Sender_hotstar	Client	TCP	15894	[TCP Previous segment not captured] https(443) → 55472 [PSH, ACK] Seq=
1067	20:23:44.517885	Sender_hotstar	Client	TLSv1.3	7254	[TCP Previous segment not captured], Continuation Data
1072	20:23:44.530399	Sender_hotstar	Client	TLSv1.3	11574	[TCP Previous segment not captured], Continuation Data
1073	20:23:44.530399	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1074	20:23:44.530399	Sender_hotstar	Client	TLSv1.3	5814	[TCP Previous segment not captured], Continuation Data
1081	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1082	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1089	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1090	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1109	20:23:44.537568	Sender_hotstar	Client	TCP	1494	[TCP Previous segment not captured] https(443) → 55472 [ACK] Seq=18075
1116	20:23:44.538033	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1117	20:23:44.538033	Sender_hotstar	Client	TLSv1.3	7254	[TCP Previous segment not captured], Continuation Data
1122	20:23:44.538280	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1123	20:23:44.538280	Sender_hotstar	Client	TLSv1.3	4374	[TCP Previous segment not captured], Continuation Data
1128	20:23:44.540597	Sender_hotstar	Client	TLSv1.3	14454	[TCP Previous segment not captured], Continuation Data
1131	20:23:44.541167	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1132	20:23:44.541167	Sender_hotstar	Client	TLSv1.3	7254	[TCP Previous segment not captured], Continuation Data
1136	20:23:44.543616	Sender_hotstar	Client	TCP	10134	[TCP Previous segment not captured] https(443) → 55472 [ACK] Seq=19188
1137	20:23:44.543616	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1138	20:23:44.543616	Sender_hotstar	Client	TLSv1.3	5814	[TCP Previous segment not captured], Continuation Data
1146	20:23:44.548435	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1147	20:23:44.548435	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1154	20:23:44.548435	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1155	20:23:44.548435	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1176	20:23:44.549256	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1177	20:23:44.549256	Sender_hotstar	Client	TLSv1.3	4374	[TCP Previous segment not captured], Continuation Data

Frame 61: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{1FA3C6B7-B8C2-4B14-8DC8-DC6DFB5525F3}, id 0

Ethernet II, Src: Client (28:cd:c4:1d:c4:0f), Dst: Gateway (30:de:4b:c7:2d:02)

Internet Protocol Version 4, Src: Client (192.168.1.102), Dst: 123.35.104.34.googleusercontent.com (34.104.35.123)

Transmission Control Protocol, Src Port: 55873 (55873), Dst Port: https (443), Seq: 0, Len: 0

Previous segment(s) not captured (common at capture start): Label

Packets: 15500 · Displayed: 598 (3.9%)

Profile: Default

lab1new_again.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.analysis.retransmission or tcp.analysis.lost_segment

No.	Time	Source	Destination	Protocol	Length	Info
10697	22:28:53.917089	clienttt	hotstarext	TCP	179	[TCP Retransmission] 55317 → https(443) [PSH, ACK] Seq=2814 Ack=1076823 Win=13
10926	22:28:54.261090	clienttt	hotstarext	TCP	490	[TCP Retransmission] 64137 → https(443) [PSH, ACK] Seq=5325 Ack=6168539 Win=26
11026	22:28:54.369559	clienttt	hotstarext	TCP	144	[TCP Retransmission] 55317 → https(443) [PSH, ACK] Seq=2884 Ack=1120651 Win=13
11402	22:28:55.069508	clienttt	hotstarext	TCP	640	[TCP Retransmission] 64137 → https(443) [PSH, ACK] Seq=5741 Ack=6350749 Win=26
12059	22:28:56.422586	clienttt	hotstarext	TCP	778	[TCP Retransmission] 64137 → https(443) [PSH, ACK] Seq=6307 Ack=6483639 Win=26
13050	22:28:58.354125	clienttt	hotstarext	TCP	215	[TCP Retransmission] 64137 → https(443) [PSH, ACK] Seq=7293 Ack=6730885 Win=26
13106	22:28:58.495366	clienttt	ma.hsr.im	TCP	144	[TCP Retransmission] 50983 → https(443) [PSH, ACK] Seq=2842 Ack=1158608 Win=65
15145	22:29:01.934910	clienttt	edge.microsoft.com	TCP	1444	[TCP Retransmission] 50985 → https(443) [ACK] Seq=2050 Ack=6423 Win=65024 Len=
15220	22:29:02.076373	clienttt	static.edge.microsoftapp...	TCP	536	[TCP Retransmission] 50984 → https(443) [PSH, ACK] Seq=2483 Ack=7111 Win=65280
15878	22:29:03.584708	clienttt	ma.hsr.im	TCP	144	[TCP Retransmission] 50983 → https(443) [PSH, ACK] Seq=2912 Ack=2053218 Win=65
23931	22:29:36.091488	clienttt	q.quora.com	TLSv1.2	397	[TCP Fast Retransmission] , Application Data, Application Data, Application Da
23966	22:29:36.352350	clienttt	hotstarext	TCP	74	[TCP Retransmission] 64137 → https(443) [FIN, ACK] Seq=12813 Ack=13330467 Win=
23967	22:29:36.352715	clienttt	www.facebook.com	TCP	74	[TCP Retransmission] 52407 → https(443) [FIN, ACK] Seq=2489 Ack=8201 Win=64512
23974	22:29:36.526588	clienttt	64:ff9b::a29f:8ce5	TCP	74	[TCP Retransmission] 65267 → https(443) [FIN, ACK] Seq=632 Ack=295 Win=255 Len
23975	22:29:36.590157	clienttt	apix.hotstar.com	TCP	74	[TCP Retransmission] 49908 → https(443) [FIN, ACK] Seq=8512 Ack=7773 Win=65280
23976	22:29:36.606196	clienttt	hotstarext	TCP	74	[TCP Retransmission] 55317 → https(443) [FIN, ACK] Seq=3304 Ack=5281190 Win=26
23985	22:29:36.669914	clienttt	64:ff9b::c7e8:649d	TCP	74	[TCP Retransmission] 62326 → https(443) [FIN, ACK] Seq=102 Ack=104 Win=254 Len
23986	22:29:36.685092	clienttt	sl-in-fi188.1e100.net	TCP	74	[TCP Retransmission] 53530 → hpvroom(5228) [FIN, ACK] Seq=2 Ack=1 Win=253 Len=
23991	22:29:36.796327	clienttt	2606:4708:8d92:7cbc:c2d9...	TCP	74	[TCP Retransmission] 53342 → https(443) [FIN, ACK] Seq=2 Ack=1 Win=253 Len=
1251	22:28:33.704410	e2c57.gcp.gvt2.com	Client	TCP	66	[TCP Retransmission] https(443) → 60607 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0
2023	22:28:35.747945	e2c57.gcp.gvt2.com	Client	TCP	66	[TCP Retransmission] https(443) → 60607 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0
2606	22:28:39.778849	e2c57.gcp.gvt2.com	Client	TCP	66	[TCP Retransmission] https(443) → 60607 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0
15658	22:29:03.084592	edge.microsoft.com	clienttt	TCP	1444	[TCP Spurious Retransmission] https(443) → 50985 [PSH, ACK] Seq=6678 Ack=17974
2523	22:28:39.309680	hotstarext	clienttt	TCP	105	[TCP Spurious Retransmission] https(443) → 64137 [PSH, ACK] Seq=4254 Ack=3087
4668	22:28:43.849838	hotstarext	clienttt	TCP	74	[TCP Retransmission] https(443) → 62129 [FIN, ACK] Seq=631 Ack=2129 Win=64128
10159	22:28:53.045508	hotstarext	clienttt	TCP	2814	[TCP Previous segment not captured] https(443) → 64137 [PSH, ACK] Seq=6171279
10166	22:28:53.046341	hotstarext	clienttt	TCP	2814	[TCP Previous segment not captured] https(443) → 64137 [PSH, ACK] Seq=6179499
10937	22:28:54.275967	hotstarext	clienttt	TCP	1444	[TCP Fast Retransmission] https(443) → 64137 [ACK] Seq=6168539 Ack=5325 Win=76
10938	22:28:54.275967	hotstarext	clienttt	TLSv1.3	1444	[TCP Fast Retransmission] , Application Data, Application Data
10947	22:28:54.276243	hotstarext	clienttt	TCP	1444	[TCP Fast Retransmission] https(443) → 64137 [ACK] Seq=6176759 Ack=5325 Win=76
11395	22:28:55.066734	o478470.ingest.sen...	clienttt	TCP	74	[TCP Retransmission] https(443) → 49234 [FIN, ACK] Seq=1 Ack=3 Win=1047 Len=0
1825	22:28:34.795187	usersvc.hotstar.com	clienttt	TCP	322	[TCP Spurious Retransmission] https(443) → 53506 [PSH, ACK] Seq=1 Ack=2113 Win
5319	22:28:45.402590	usersvc.hotstar.com	clienttt	TCP	74	[TCP Retransmission] https(443) → 53506 [FIN, ACK] Seq=599 Ack=2177 Win=69760

..... 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)

Previous segment(s) not captured (common at capture start): Label

Packets: 24009 - Displayed: 81 (0.3%)

Profile: Default

Session (time)	Throughput (bps)	RTT (max ms)	Avg Pkt Size (B) , Count	Loss (%)	#TCP
Afternoon	5176 k	50	2178 , 15500	3.9	3503 (24 per of total)
Night	2728 k	40	1151, 24009	0.3	15351 (30 per of total)

6 Content Source(s) (Q f)

Streaming services like Hotstar rely on CDNs (Content Delivery Networks) to ensure smooth delivery of live content.

Reasoning

- Various addresses during resolution:

```
▼ Domain Name System (response)
  Transaction ID: 0x51cf
  ▶ Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 1
  Authority RRs: 4
  Additional RRs: 2
  ▼ Queries
    ▶ aba5c6eaaaaaamon3ym4l56kplk.live-cf.cdn.hotstar.com: type A, class IN
  ▼ Answers
    ▶ aba5c6eaaaaaamon3ym4l56kplk.live-cf.cdn.hotstar.com: type A, class IN, addr 70.130.228.70

▼ Domain Name System (response)
  Transaction ID: 0x81df
  ▶ Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 11
  Authority RRs: 8
  Additional RRs: 1
  ▼ Queries
    ▶ bifrost-api.hotstar.com: type A, class IN
  ▼ Answers
    ▶ bifrost-api.hotstar.com: type CNAME, class IN, cname bifrost-api.hotstar.com-api7.edgekey.net
    ▶ bifrost-api.hotstar.com-api7.edgekey.net: type CNAME, class IN, cname e35862.dscapi7.akamaiedge.net
    ▶ e35862.dscapi7.akamaiedge.net: type A, class IN, addr 23.213.12.211
    ▶ e35862.dscapi7.akamaiedge.net: type A, class IN, addr 23.213.12.139
    ▶ e35862.dscapi7.akamaiedge.net: type A, class IN, addr 23.213.12.232
    ▶ e35862.dscapi7.akamaiedge.net: type A, class IN, addr 23.213.12.194
    ▶ e35862.dscapi7.akamaiedge.net: type A, class IN, addr 23.213.12.147
    ▶ e35862.dscapi7.akamaiedge.net: type A, class IN, addr 23.213.12.171
    ▶ e35862.dscapi7.akamaiedge.net: type A, class IN, addr 23.213.12.179
    ▶ e35862.dscapi7.akamaiedge.net: type A, class IN, addr 23.213.12.184
    ▶ e35862.dscapi7.akamaiedge.net: type A, class IN, addr 23.213.12.162
  ▼ Authoritative nameservers
    ▶ dscapi7.akamaiedge.net: type NS, class IN, ns n5dscapi7.akamaiedge.net
    ▶ dscapi7.akamaiedge.net: type NS, class IN, ns n3dscapi7.akamaiedge.net
    ▶ dscapi7.akamaiedge.net: type NS, class IN, ns n4dscapi7.akamaiedge.net
    ▶ dscapi7.akamaiedge.net: type NS, class IN, ns n6dscapi7.akamaiedge.net
```

(g) HTTP GET and POST Transactions Analysis

During the analysis of Hotstar live sports streaming traffic in Wireshark, we observed that **all application traffic was encrypted using HTTPS over TLS (port 443)**. As a result, plain HTTP messages (GET/POST) were not visible in the packet traces, and only TLS Handshake and Application Data records appeared.

To address this, we cross-verified with browser developer tools (Chrome/Edge Network Tab), which revealed the use of:

- **HTTP GET** requests: used to fetch video resources). These correspond to frequent sequential requests seen in the TLS streams.
- **HTTP POST** requests: used to transmit authentication tokens, session keep-alive messages. These appear less frequently but with larger payloads.

Observed Parameters from TLS Traces

- **Protocols:** TCP/IPv4 (Application data encrypted)
- **Destination Port:** 55472
- **source Port:** Https 443

```

Transmission Control Protocol, Src Port: https (443), Dst Port: 55472 (55472), Seq: 1771960, Ack: 4975, Len: 1440
  Source Port: https (443)
  Destination Port: 55472 (55472)
  [Stream index: 16]
  [Stream Packet Number: 840]
  [Conversation completeness: Incomplete, DATA (15)]
  [TCP Segment Len: 1440]
  Sequence Number: 1771960 (relative sequence number)
  Sequence Number (raw): 2122752491
  [Next Sequence Number: 1773400 (relative sequence number)]
  Acknowledgment Number: 4975 (relative ack number)
  Acknowledgment number (raw): 134479340
  0101 .... = Header Length: 20 bytes (5)
  [Flags: 0x010 (ACK)]
  Window: 158
  [Calculated window size: 80896]
  [Window size scaling factor: 512]
  Checksum: 0x25af [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  [Timestamps]
  TCP payload (1440 bytes)
Transport Layer Security

```

Explanation of HTTP GET and POST

HTTP GET:

- Purpose: Retrieve data from the server (idempotent, safe).
- Fields of interest:
 - **Host:** Domain name of the server.
 - **User-Agent:** Identifies browser or application.
 - **Accept:** Specifies accepted response formats (e.g., video/mp4, application/json).

HTTP POST:

- Purpose: Send data to the server (e.g., authentication, form submission).
- Fields of interest:
 - **Host:** Server domain.
 - **Content-Type:** Format of the data in body (application/json, application/x-www-form-urlencoded).
 - **Content-Length:** Size of the message body in bytes.
 - **Body:** Actual data payload (e.g., JSON object with session info).

Snapshots

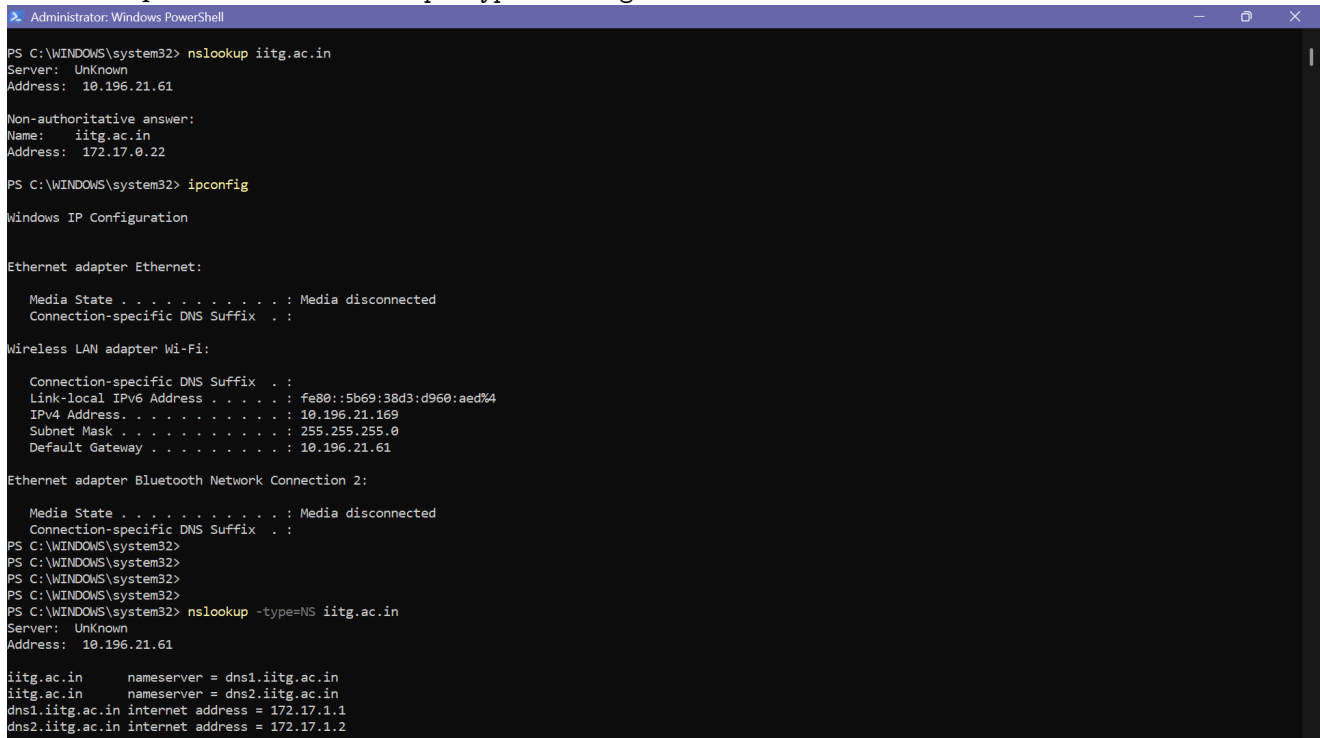
1069	20:23:44.527675	Sender_hotstar	Client	TLSv1.3	1494	Continuation Data
1071	20:23:44.530399	Sender_hotstar	Client	TLSv1.3	2934	Continuation Data
1072	20:23:44.530399	Sender_hotstar	Client	TLSv1.3	11574	[TCP Previous segment not captured], Continuation Data
1073	20:23:44.530399	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1074	20:23:44.530399	Sender_hotstar	Client	TLSv1.3	5814	[TCP Previous segment not captured], Continuation Data
1079	20:23:44.531519	Sender_hotstar	Client	TLSv1.3	8694	Continuation Data
1081	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1082	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1083	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	Continuation Data
1084	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	Continuation Data
1085	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	Continuation Data
1086	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	Continuation Data
1087	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	Continuation Data
1088	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	Continuation Data
1089	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1090	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1091	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	Continuation Data
1092	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	1494	Continuation Data
1093	20:23:44.534311	Sender_hotstar	Client	TLSv1.3	4374	Continuation Data
1107	20:23:44.534881	Sender_hotstar	Client	TLSv1.3	2934	Continuation Data
1115	20:23:44.538033	Sender_hotstar	Client	SSLv2	10134	Encrypted Data, Continuation Data
1116	20:23:44.538033	Sender_hotstar	Client	TLSv1.3	1494	[TCP Previous segment not captured], Continuation Data
1117	20:23:44.538033	Sender_hotstar	Client	TLSv1.3	7254	[TCP Previous segment not captured], Continuation Data

(h) DNS Analysis for IIT Guwahati

We used both nslookup and Wireshark to analyze DNS queries related to IIT Guwahati.

Findings from nslookup

- **Resolved IP Address:** 172.17.0.22
Server: UnKnown
Address: 10.196.21.61 iitg.ac.in
nameserver = dns1.iitg.ac.in iitg.ac.in
nameserver = dns2.iitg.ac.in dns1.iitg.ac.in internet address = 172.17.1.1
dns2.iitg.ac.in internet address = 172.17.1.2
- **NS Lookup Command:** nslookup -type=NS iitg.ac.in



```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> nslookup iitg.ac.in
Server: UnKnown
Address: 10.196.21.61

Non-authoritative answer:
Name: iitg.ac.in
Address: 172.17.0.22

PS C:\WINDOWS\system32> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::5b69:38d3:d960:aed%4
    IPv4 Address. . . . . : 10.196.21.169
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.196.21.61

Ethernet adapter Bluetooth Network Connection 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> nslookup -type=NS iitg.ac.in
Server: UnKnown
Address: 10.196.21.61

iitg.ac.in    nameserver = dns1.iitg.ac.in
iitg.ac.in    nameserver = dns2.iitg.ac.in
dns1.iitg.ac.in internet address = 172.17.1.1
dns2.iitg.ac.in internet address = 172.17.1.2
```

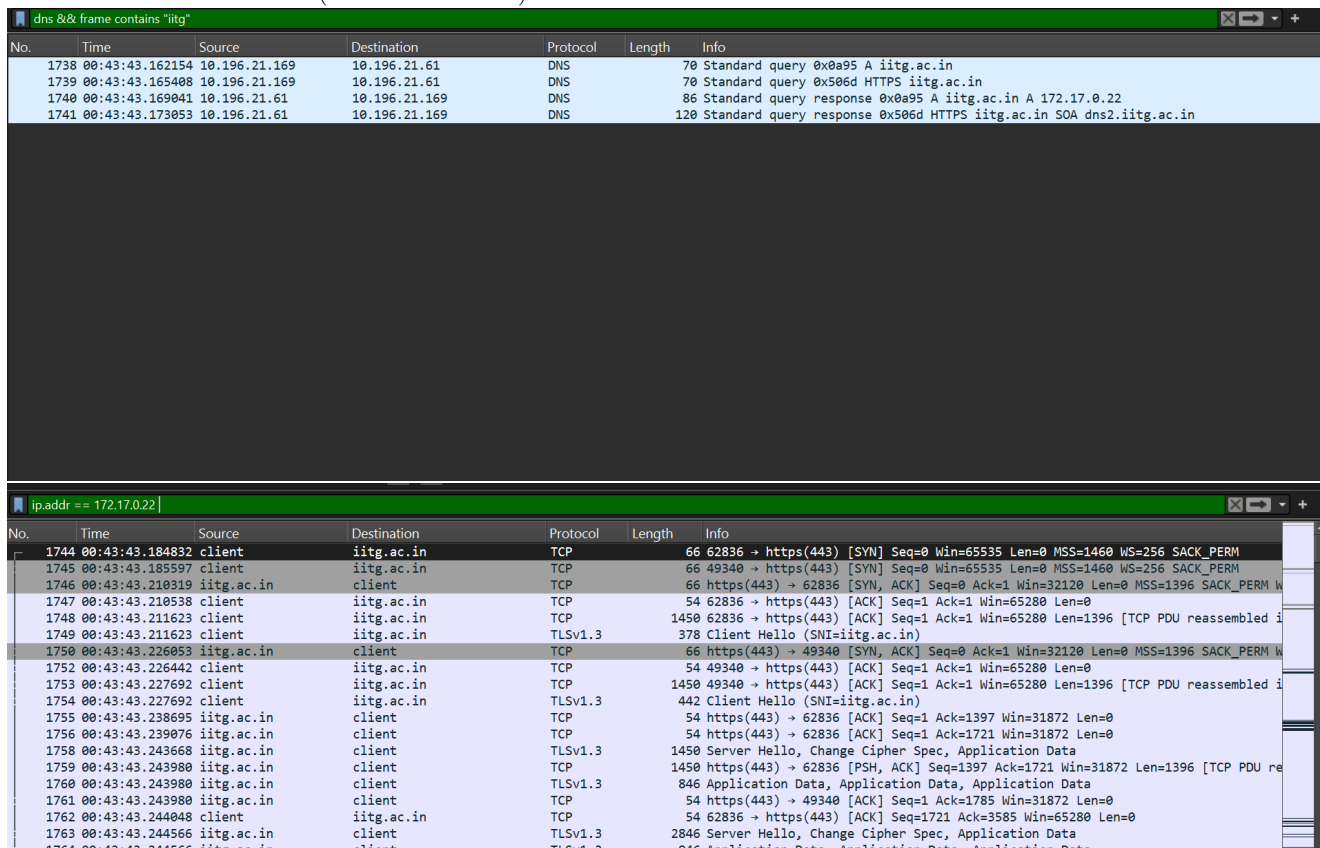
Server: UnKnown
Address: 10.196.21.61

Name: iitg.ac.in
Address: 172.17.0.22

PS C:\WINDOWS\system32>

Wireshark DNS Trace Analysis

- Source IP : 172.17.0.22
- Number of Questions: 1 (for www.iitg.ac.in)
- Number of Answers: 1 (with resolved IP)



The image shows two Wireshark packet capture windows. The top window is filtered for 'dns && frame contains "iitg"', showing four DNS packets. The bottom window is filtered for 'ip.addr == 172.17.0.22', showing a full TCP/TLS session for a request to iitg.ac.in.

No.	Time	Source	Destination	Protocol	Length	Info
1738	00:43:43.162154	10.196.21.169	10.196.21.61	DNS	70	Standard query 0x0a95 A iitg.ac.in
1739	00:43:43.165408	10.196.21.169	10.196.21.61	DNS	70	Standard query 0x506d HTTPS iitg.ac.in
1740	00:43:43.169041	10.196.21.61	10.196.21.169	DNS	86	Standard query response 0x0a95 A iitg.ac.in A 172.17.0.22
1741	00:43:43.173053	10.196.21.61	10.196.21.169	DNS	120	Standard query response 0x506d HTTPS iitg.ac.in SOA dns2.iitg.ac.in

No.	Time	Source	Destination	Protocol	Length	Info
1744	00:43:43.184832	client	iitg.ac.in	TCP	66	62836 → https(443) [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1745	00:43:43.185597	client	iitg.ac.in	TCP	66	49340 → https(443) [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1746	00:43:43.210319	iitg.ac.in	client	TCP	66	https(443) → 62836 [SYN, ACK] Seq=0 Ack=1 Win=32120 Len=0 MSS=1396 SACK_PERM
1747	00:43:43.210538	client	iitg.ac.in	TCP	54	62836 → https(443) [ACK] Seq=1 Ack=1 Win=65280 Len=0
1748	00:43:43.211623	client	iitg.ac.in	TCP	1450	62836 → https(443) [ACK] Seq=1 Ack=1 Win=65280 Len=1396 [TCP PDU reassembled i
1749	00:43:43.211623	client	iitg.ac.in	TLSv1.3	378	Client Hello (SNI=iitg.ac.in)
1750	00:43:43.226053	iitg.ac.in	client	TCP	66	https(443) → 49340 [SYN, ACK] Seq=0 Ack=1 Win=32120 Len=0 MSS=1396 SACK_PERM
1752	00:43:43.226442	client	iitg.ac.in	TCP	54	49340 → https(443) [ACK] Seq=1 Ack=1 Win=65280 Len=0
1753	00:43:43.227692	client	iitg.ac.in	TCP	1450	49340 → https(443) [ACK] Seq=1 Ack=1 Win=65280 Len=1396 [TCP PDU reassembled i
1754	00:43:43.227692	client	iitg.ac.in	TLSv1.3	442	Client Hello (SNI=iitg.ac.in)
1755	00:43:43.238695	iitg.ac.in	client	TCP	54	https(443) → 62836 [ACK] Seq=1 Ack=1397 Win=31872 Len=0
1756	00:43:43.239076	iitg.ac.in	client	TCP	54	https(443) → 62836 [ACK] Seq=1 Ack=1721 Win=31872 Len=0
1758	00:43:43.243668	iitg.ac.in	client	TLSv1.3	1450	Server Hello, Change Cipher Spec, Application Data
1759	00:43:43.243980	iitg.ac.in	client	TCP	1450	https(443) → 62836 [PSH, ACK] Seq=1397 Ack=1721 Win=31872 Len=1396 [TCP PDU re
1760	00:43:43.243980	iitg.ac.in	client	TLSv1.3	846	Application Data, Application Data, Application Data
1761	00:43:43.243980	iitg.ac.in	client	TCP	54	https(443) → 49340 [ACK] Seq=1 Ack=1785 Win=31872 Len=0
1762	00:43:43.244048	client	iitg.ac.in	TCP	54	62836 → https(443) [ACK] Seq=1721 Ack=3585 Win=65280 Len=0
1763	00:43:43.244566	iitg.ac.in	client	TLSv1.3	2846	Server Hello, Change Cipher Spec, Application Data

/

Caching Observation

When accessing www.iitg.ac.in multiple times, we observed that subsequent requests did not generate new DNS queries in Wireshark. This confirms that DNS caching occurred, either at the client OS or the resolver.